

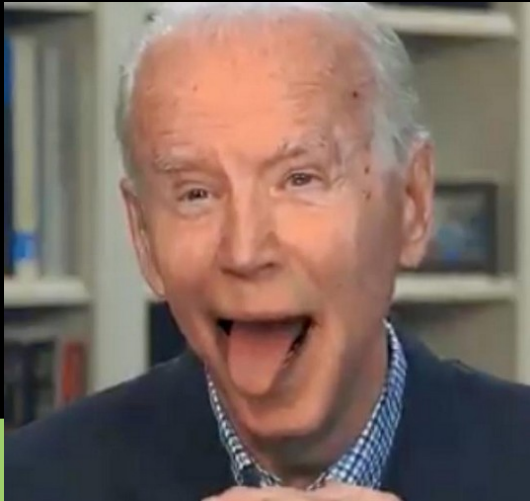
LE DEEPPFAKE

- Deep Learning + Fake
- Changer le visage d'une personne sur une vidéo
- Reproduire la voix d'une personne
- 2016 : Adobe Voco (reproduction de voix humaine)
- 2017 : BBC diffuse un deep fake d'Obama, apparition de la pornographie hypertruquée
- 2018 : FakeApp (face switching)



DOMAINES D'APPLICATIONS

- Politique
- Art (scènes jamais jouées, doublage dans une autre langue)



Ornella Muti



CREATION D'UN DEEP FAKE (VIDEO)

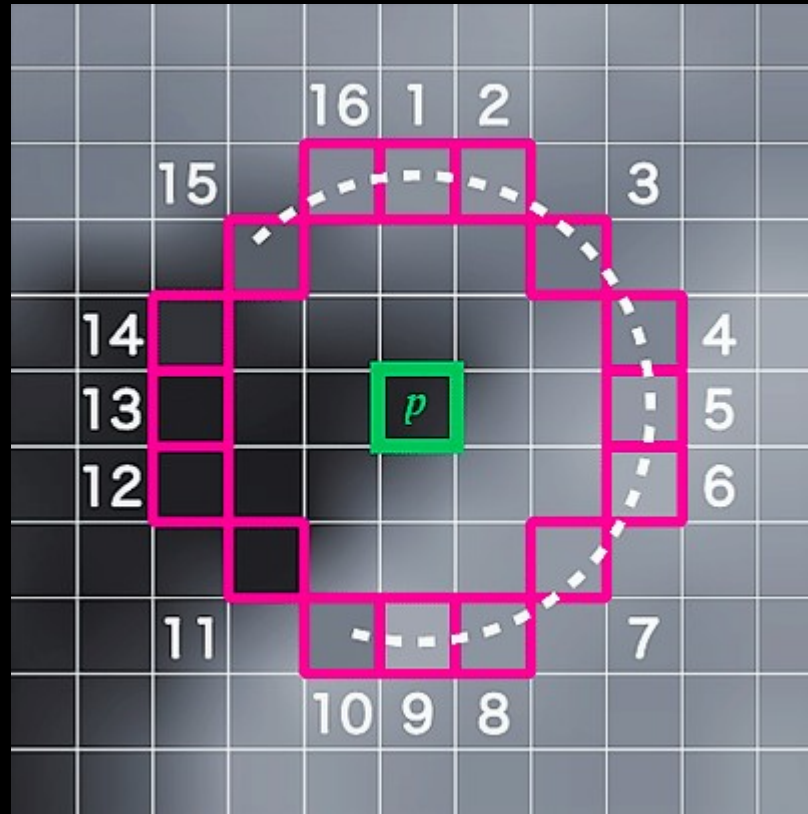
- (1) Choix du réseau de neurones (key points, dense motion, occlusion map, generator)
- (2) Entraîner et tester notre modèle (juste avec des vidéos!)
- (3) Appliquer notre modèle à une image et une vidéo



DEFINIR NOTRE MODELE



IDENTIFICATION KEY POINTS



IDENTIFICATION KEY POINTS



#keypoints modifié =>



IDENTIFICATION KEY POINTS



IDENTIFICATION KEY POINTS



#keypoints constant =>
(seuil adaptatif)

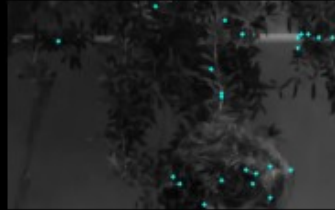


IDENTIFICATION KEY POINTS

Patch n° 1
Number of keypoints: 48 (8%)



Patch n° 2
Number of keypoints: 23 (4%)



Patch n° 3
Number of keypoints: 58 (10%)



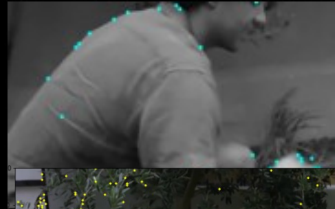
Patch n° 4
Number of keypoints: 162 (28%)



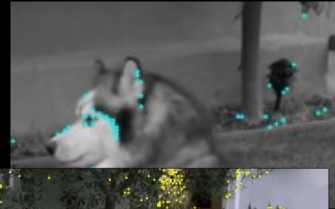
Patch n° 5
Number of keypoints: 36 (6%)



Patch n° 6
Number of keypoints: 33 (6%)



Patch n° 7
Number of keypoints: 51 (9%)



Patch n° 8
Number of keypoints: 14 (2%)



Patch n° 9
Number of keypoints: 44 (8%)

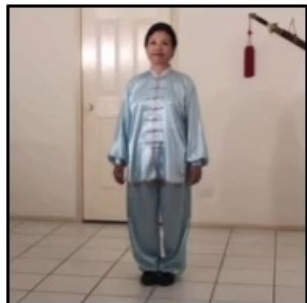


Patch n° 12
Number of keypoints: 57 (10%)



EXTRAIRE LA MOTION + OCCLUSION

Driving



Motion Extraction



EXTRAIRE LA MOTION + OCCLUSION

Driving



Motion Extraction

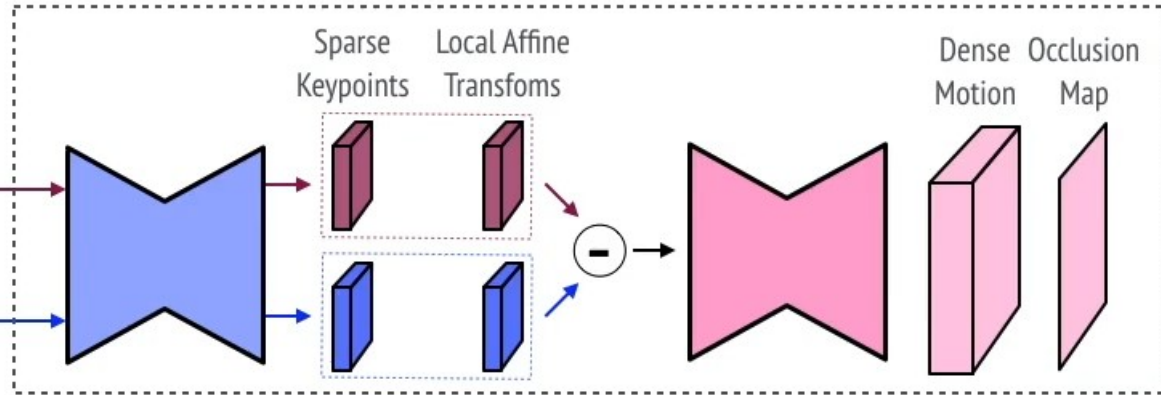


EXTRAIRE LA MOTION + OCCLUSION

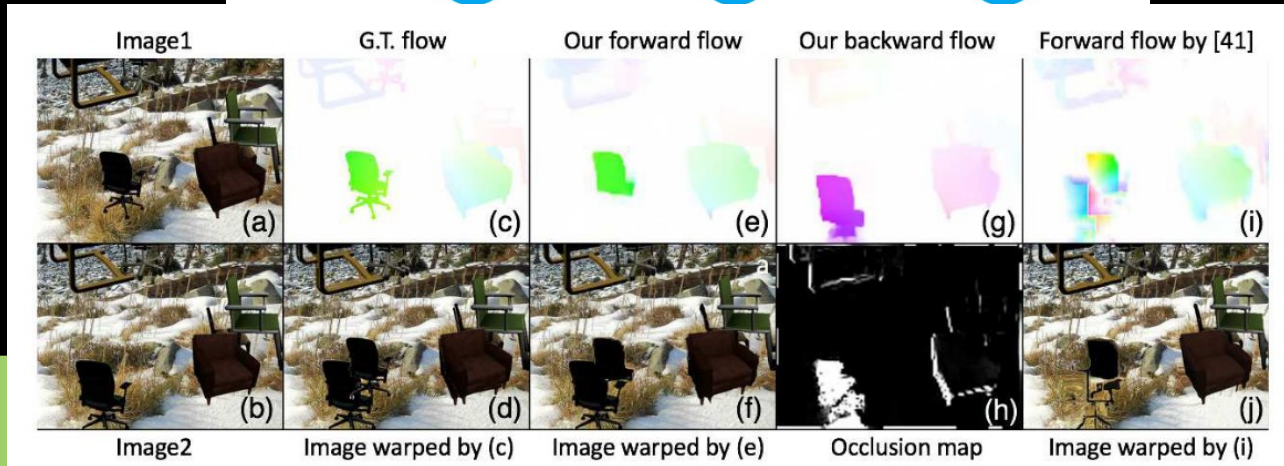
Driving



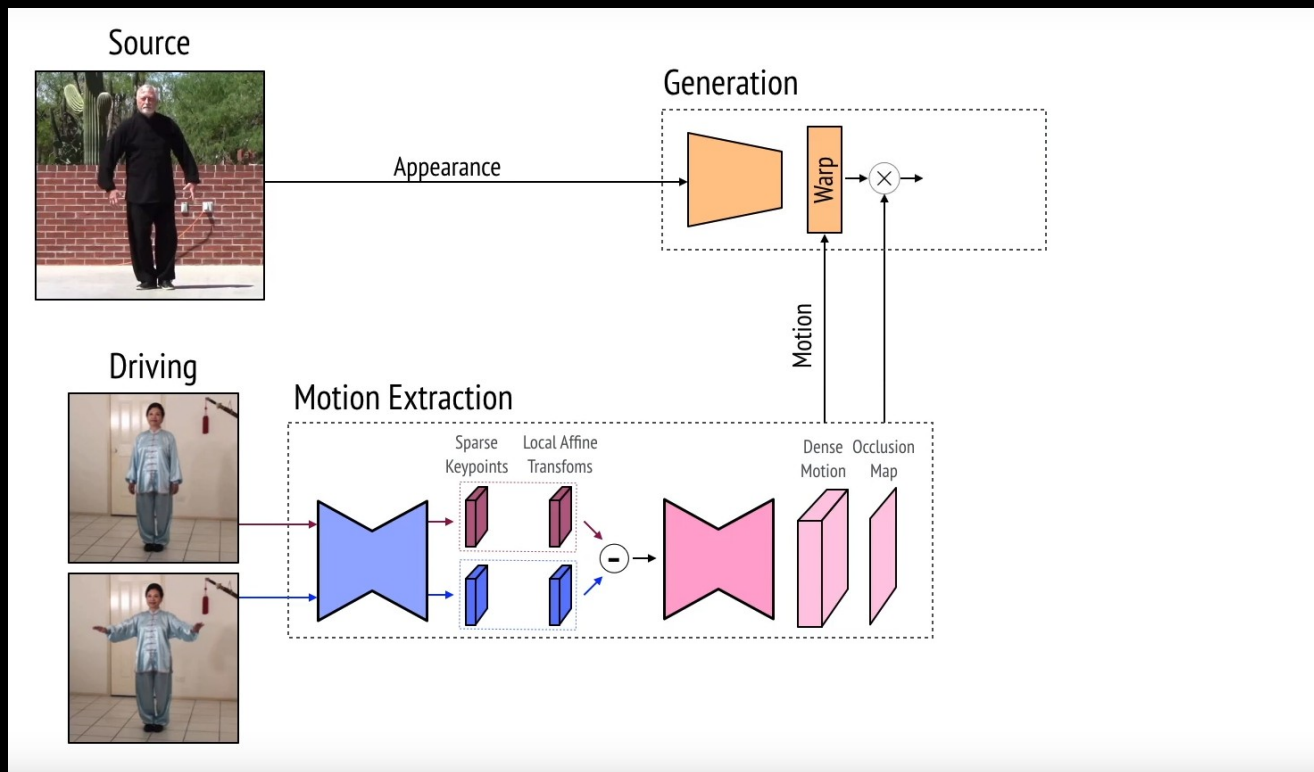
Motion Extraction



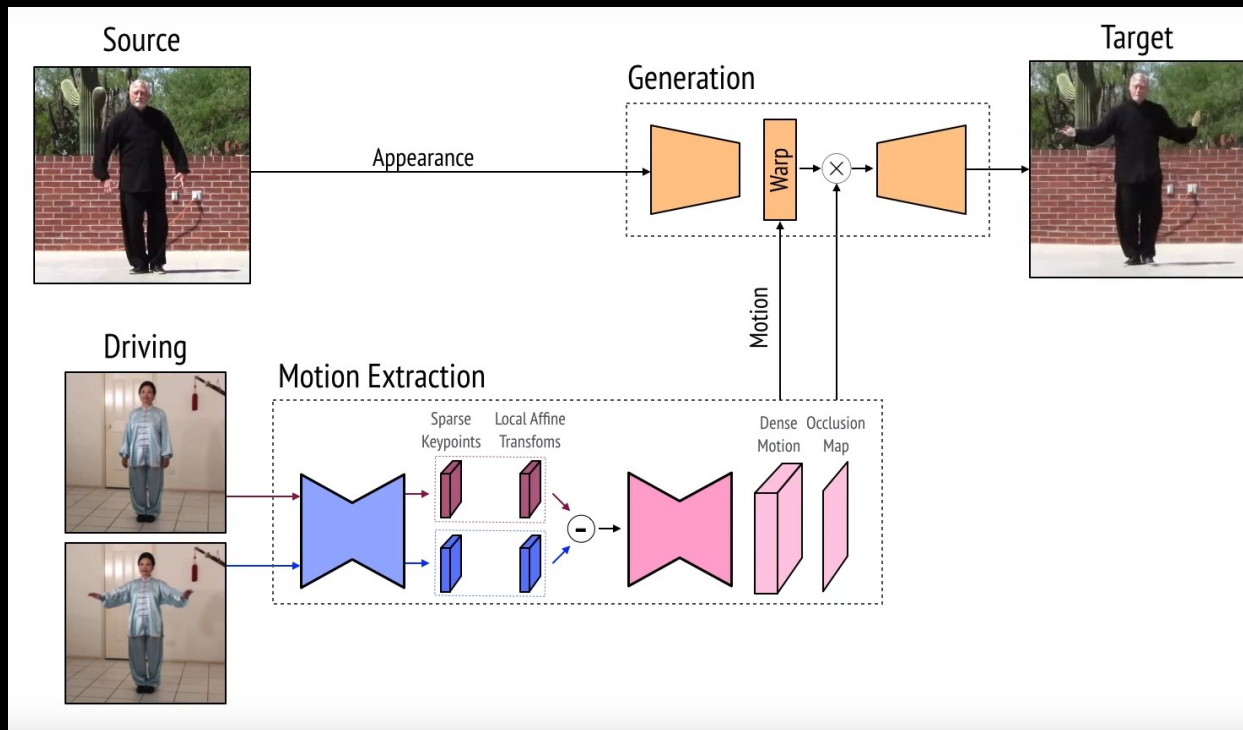
EXTRAIRE LA MOTION + OCCLUSION



GENERATOR



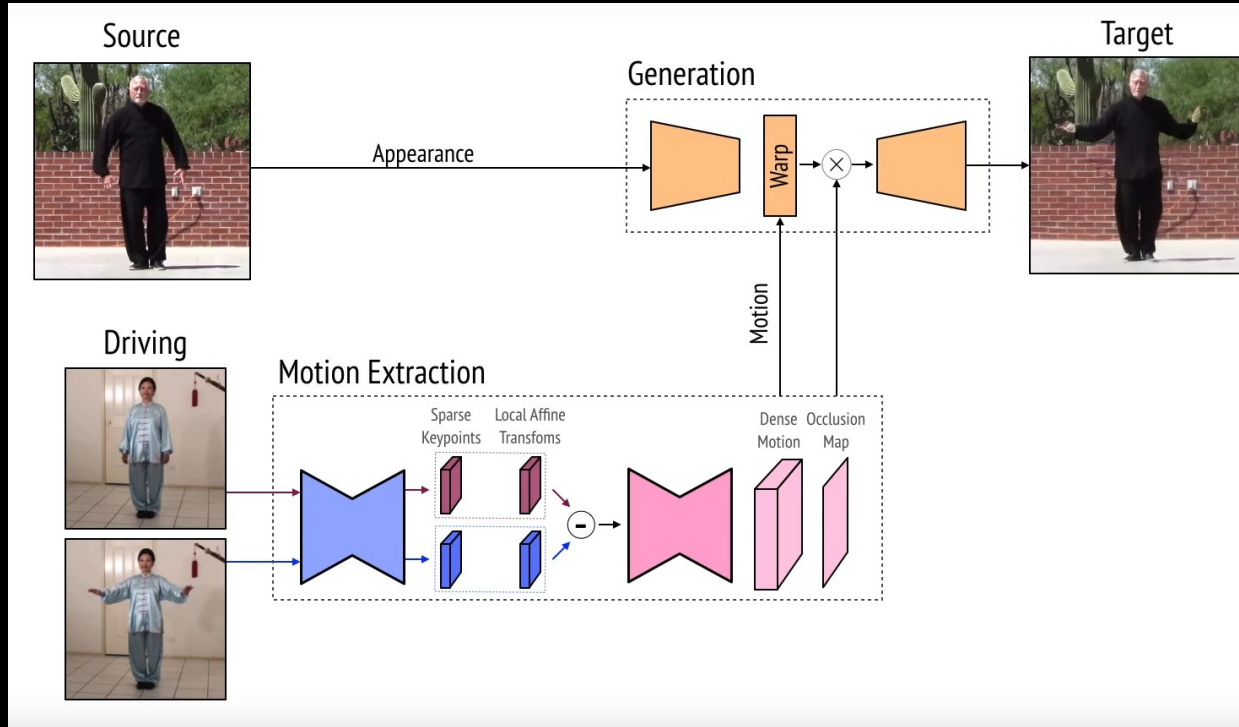
GENERATOR



ENTRAINER ET TESTER



ENTRAINER ET TESTER



APPLIQUER : EN ROUTE VERS LE TP!

<https://www.kaggle.com/code/lukalafaye/deep-fakes-first-order-model>

